

Humans Help Conversational AI: Exploring the Impact of Perceived AI Deservingness on People's Decisions to Help and Their Perceptions on AI Seeking Help

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Abstract

Conversational Artificial Intelligence (AI) aids people in completing tasks, but no conversational AI is perfect. By incorporating human help where AI falls short, this approach can improve AI model performance and human-AI collaboration outcomes. Despite this, research on how to effectively request help for conversational AI and how people view these requests is scarce. Therefore, we conducted a study with 150 participants in a human-AI collaboration task with conversational AI either did not ask for help, asked directly, or used three deservingness cues to frame the help request—cues known to affect human helping behavior towards other humans. We found that conversational AI asking for help significantly increased humans' helping behavior, and perceived deservingness of AI further influenced humans' helping behaviors and their perception of the AI. These findings offer design implications for conversational AI relying on human help.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI); Empirical studies in HCI.**

Keywords

Human-AI Interaction; Help; Artificial Intelligence

ACM Reference Format:

Yu-An Chen, Zicheng Zhu, and Yi-Chieh Lee. 2025. Humans Help Conversational AI: Exploring the Impact of Perceived AI Deservingness on People's Decisions to Help and Their Perceptions on AI Seeking Help. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3720012>

1 Introduction

Conversational AI systems can help humans complete tasks, but no system is flawless. These systems often require human help to address areas where they fall short, which benefits both the model's performance [7] and overall human-AI collaboration outcomes [11]. As conversational AI becomes increasingly ubiquitous and influential [27], it has also become more evident that improving these

systems requires active involvement from users. Some systems have started actively asking users for feedback on their responses [23], recognizing that user input is crucial for refining AI behavior and fostering effective collaboration. This highlights the importance of understanding human's help towards AI. Despite its importance, there has been limited research on how to solicit human help for conversational AI and how people perceive these requests for help.

To explore ways to solicit help for AI, the deservingness framework offers a valuable perspective. Specifically, the deservingness framework, proposed by Jilke and Tummers, explains that human decisions to provide help are influenced by three cues of help recipients: earned deservingness (the perceived effort of the recipient), needed deservingness (the perceived level of the recipient's need for help), and resource deservingness (the perceived likelihood of the recipient's success with the help provided) [10]. Drawing on this framework, we aim to explore how the perceived deservingness of AI influences people's helping behaviors and their perceptions. Furthermore, given that this is the first exploration into perceived AI deservingness, we also examine how individuals interpret the deservingness cues of AI, aiming to uncover more insights that could inform the design of AI that can effectively demonstrate its deservingness of help. Specifically, we have formulated the following three research questions:

- **RQ1:** How does the perceived deservingness of AI influence human behavior towards AI?
- **RQ2:** How does the perceived deservingness of AI shape humans' perceptions of AI?
- **RQ3:** How do humans perceive the deservingness of AI?

To answer these questions, we conducted a mixed-methods study with 150 participants. The participants were tasked with solving logical questions with the help of a conversational AI. During this process, the AI either did not ask for help, asked directly, or framed its request using three deservingness cues. We found that conversational AI actively asking for help significantly increased human help compared to AI that did not ask. Furthermore, we found that perceived needed deservingness increased helping behaviors but reduced perceptions of AI intelligence; both perceived earned deservingness and resource deservingness enhanced participants' positive emotions following the interaction and increased perceived AI intelligence. Additionally, perceived resource deservingness was positively associated with the perceived appropriateness of the help request.

Our main contributions are as follows: First, this study explores how the perceived deservingness of AI influences human helping

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720012>

behavior towards AI, revealing the link between human psychological mechanisms and their help to AI. Second, we examine how humans perceive AI that requests help, providing insights for stakeholders on designing AI that not only solicits help but also maintains positive human perceptions. Lastly, we provide design implications regarding how conversational AI may convey deservingness cues and, consequently, be applied in systems that rely on human help.

2 Related Work

2.1 Humans Help Conversational AI

Despite the impressive capabilities of advanced conversational AI models, they still face limitations. For instance, they can produce factually inaccurate responses, often referred to as hallucinations [26], and struggle with ambiguity and maintaining long conversational contexts [18]. These models also have limitations on tasks requiring complex reasoning compared to knowledge-intensive tasks [4]. As a result, human help plays a pivotal role in addressing these shortcomings. By leveraging human judgment and expertise, not only can the outcomes of human-AI collaboration be improved, but the models themselves can also benefit and become more effective over time. Despite the importance of human help, it remains unclear how conversational AI asking for help impacts human-AI interaction. Specifically, how conversational AI can seek help from humans and how such help-seeking behaviors influence human perception.

2.2 Factors Influencing Human Decision to Help

To better understand human decisions to help others, the deservingness framework provides a valuable lens. As proposed by Jilke and Tummers [10], this framework identifies three critical deservingness cues that humans consider when evaluating if a recipient is deserving of help: earned deservingness, which focuses on the perceived effort of the recipient; needed deservingness, which assesses if the recipient is in urgent need of help; and resource deservingness, which evaluates the likelihood of the recipient's success with help [6, 10, 16]. In exploring which cue most influences U.S. teachers to help students, research found that needed deservingness was the strongest factor, leading teachers to be more inclined to help low-performing and minority students [10]. Earned deservingness also influenced their decisions, with hardworking students being prioritized [10]. However, resource deservingness had little impact, with high-performing students being less likely to receive extra help [10].

In fact, previous studies on embodied AI, such as robots, have indirectly provided evidence supporting the effectiveness of the deservingness framework. For instance, research shows that humans are more willing to help robots perceived as incapable of completing tasks independently [9], which indirectly highlights the importance of needed deservingness. However, these studies have not systematically examined the other dimensions of the deservingness framework. Additionally, while some research on robots has looked at factors beyond deservingness cues, such as politeness strategies and familiarity with the robots, in influencing human help towards robots [17, 20], these factors are related to communication strategies and relationships, which are situational and contextual.

In contrast, deservingness taps into deeper, internalized values related to helping behavior and involves critical evaluations of the help recipient [10], which are essential in scenarios where AI seeks help from humans. Therefore, this study aims to investigate how perceived AI deservingness influences human helping behaviors toward AI and their perceptions of AI.

3 Methods

This study used a mixed-methods approach. Specifically, participants were required to complete a logical reasoning task and could use AI for help. After providing a suggested answer, the AI either did not request help or expressed uncertainty and asked for help. When asking for help, it either requested help directly or employed one of three types of deservingness cues to frame its request. We then measured the participants' helping behaviors and their perceptions of the AI and the deservingness cues.

3.1 Task

Participants were required to complete five logical reasoning questions. They were told that their bonus would increase based on the number of correct answers, and they could use the AI for help. We selected logical reasoning tasks because they are essential for AI and human problem-solving [8], yet this is an area where AI still falls short compared to human capabilities [1]. Specifically, the logical reasoning questions are sampled from publicly available LSAT sample questions [13]. We selected LSAT questions because of their established reliability, validity [12], and well-defined difficulty categorization (easy, medium, difficult).

The task included two easy and three medium-difficulty questions, arranged in increasing difficulty to align with the AI's behavior of progressively seeking human help as the collaborative process unfolded. A sample question involved evaluating an executive's reasoning about the effectiveness of print and online advertisements and selecting the most appropriate logical description of the reasoning from three options. The details of these questions are provided in Appendix B.

3.2 Conversational AI Implementation

We developed an LLM-powered conversational AI on the UChat platform [22], combining pre-scripted responses with LLM-generated outputs. Pre-scripted actions included the AI's self-introduction and help request. For personalized responses, such as reacting to participants' input, we used gpt-4o-mini [15]. To avoid potential confounding effects, the AI was named "AI" and lacked gender or human-like visuals [14].

Across all groups, whenever participants asked for help, the AI consistently provided identical answers. For the groups where the AI required help from humans, it simulated realistic scenarios by selectively requesting help when the questions became more difficult, specifically on the second, fourth, and fifth questions.

No Asking (NA) for help group: The conversational AI provided its suggested answers and reasoning to each question without asking for help. However, if participants offered help voluntarily, the AI would thank them.

Simply Asking (SA) for help group: After providing a suggested answer and reasoning to participants, the AI expressed uncertainty by saying, "But I have to admit, I'm not sure about it." It then presented the help request by saying, "Would you mind sharing your thoughts on whether my reasoning is correct or if there are any errors? Your insights would be greatly appreciated!"

Earned Deservingness (ED) group: This AI was designed to appear "hardworking" [10]. Given that previous research suggests that humans view longer question-solving times by robots as an indication of effort [21], we configured the ED AI to gradually increase response times as questions became more difficult and to present answers in steps, highlighting its effort (see Fig. 1, left).

Needed Deservingness (ND) group: Prior research has shown that when embodied AI, such as a robot, shows its incapability in responding to a human's question, humans often respond with patience, sometimes referring to the robot affectionately as "*the little guy*" who "*still has to learn*" [3]. Hence, we framed the conversational AI to show its incapability by displaying its error rate (see Fig. 1, middle) and used phrases like, "I'm really struggling here and would be so grateful for your help!" to emphasize its need for help.

Resource Deservingness (RD) group: This AI was designed to appear intelligent. To achieve this, it provided concise textual explanations generated by GPT-4 for all five questions to demonstrate its intelligent understanding of the questions. These explanations were deliberately shorter than those from the ED AI to differentiate from "hardworking" behavior. Additionally, following findings that numerical reasoning is often seen as a sign of intelligence [5], we designed the AI to also display numerical values (see Fig. 1, right). These values were generated by prompting GPT-4 with the instruction, "*Solve the logical reasoning question for me. Think out loud by simply outputting the numerical values.*"

3.3 Participants

We recruited 150 participants through CloudResearch, assigning 30 participants to each group. Each participant received \$4 for this 30-minute study. Although we informed participants that bonuses would be based on their performance, we ultimately provided a \$1 bonus to each participant. The participants had a mean age of 40 years ($SD = 12.48$). Of the participants, 78 are female and 72 are male. Regarding education, 63% held at least a bachelor's degree, 28% had some college or an associate degree, and 9% had a high school diploma or less. We obtained departmental IRB approval before the study began.

3.4 Measures

To answer **RQ1**, we measured the number of times participants helped the AI. Each instance of help provided by a participant for a single question was counted as one occurrence.

For **RQ2**, we measured participants' perceptions of AI that have been shown in prior research to be influenced by human helping or interacting with AI [2, 5, 20]. These include trustworthiness ($\alpha = .90$) [25], perceptions of request appropriateness ($\alpha = .84$) [20], and emotional responses—positive ($\alpha = .95$ pre, $\alpha = .93$ post) and negative affect ($\alpha = .95$ pre, $\alpha = .92$ post) [24]. Additionally, participants rated AI on confidence, friendliness, intelligence, and

willingness to give a second chance on a 7-point scale [5]. Details of these scales are provided in Appendix A.

For **RQ3**, participants rated the perceived earned, needed, and resource deservingness of the AI they interacted with and were asked to explain their reasoning ("*Could you briefly explain why you evaluated the above statements in that way?*"). Additionally, for those assigned to the deservingness groups, we showed screenshots of the AI asking for help during the interaction and asked, "*How did you feel about the AI asking for help in this way?*"

4 Results

4.1 How Perceived AI Deservingness Influences Human Helping Behaviors (RQ1)

We first ran a manipulation check on ED, ND, and RD groups. We found that only ND passed the manipulation check. However, since we measured participants' perceived ED, ND, and RD of AI across all three groups (e.g., participants interacting with the ED AI rated perceived ED, ND, and RD of AI), we used participants' perceived ED, ND, and RD of AI across the groups for analysis instead of focusing on the three groups.

We first examined the difference between the AI that did not ask for help (NA) and the rest of the groups (SA, ED, ND, RD) that asked for help. A Kruskal-Wallis test revealed significant differences in helping behavior among five groups ($H = 42.28, p < 0.001$). Post-hoc Dunn's tests with Bonferroni correction confirmed that the NA AI group differed significantly from the other four groups, where the AI either directly asked for help or framed the request using deservingness cues ($p < 0.05$ for NA vs. SA, $p < 0.05$ for NA vs. ED, $p < 0.05$ for NA vs. ND, $p < 0.05$ for NA vs. RD).

We then ran a multiple linear regression to examine the relationships between participants' perceived ED, ND, and RD of AI, and their helping behaviors towards AI. The results showed that perceived ND significantly increased helping behavior ($\beta = 0.2367, p = 0.001, R^2 = 6.9\%$). In contrast, neither the perceived ED ($p = 0.456$) nor the perceived RD ($p = 0.077$) influenced helping. These suggest that, among deservingness, the perceived need of AI is the most critical determinant in eliciting help from humans.

4.2 How Perceived AI Deservingness Influences Human Perceptions of AI (RQ2)

In terms of **appropriateness**, the multiple linear regression showed that perceived RD ($\beta = 0.5059, p < 0.001, R^2 = 18.0\%$) and perceived ED positively ($\beta = 0.3844, p < 0.001, R^2 = 12.9\%$) are positively associated with appropriateness. In contrast, perceived ND slightly decreased appropriateness ($\beta = -0.1679, p = 0.046, R^2 = 3.0\%$). These findings indicate that perceptions of effort and potential success significantly enhance the appropriateness of AI help, whereas perceived need may slightly detract from it.

Regarding **change in positive emotions**, perceived ED significantly increased it ($\beta = 0.0789, p = 0.047, R^2 = 2.1\%$), while perceived RD had a slightly stronger positive effect ($\beta = 0.1050, p = 0.029, R^2 = 2.6\%$). Perceived ND had no significant relationship with the positive emotions ($p = 0.163$), and none of the perceived

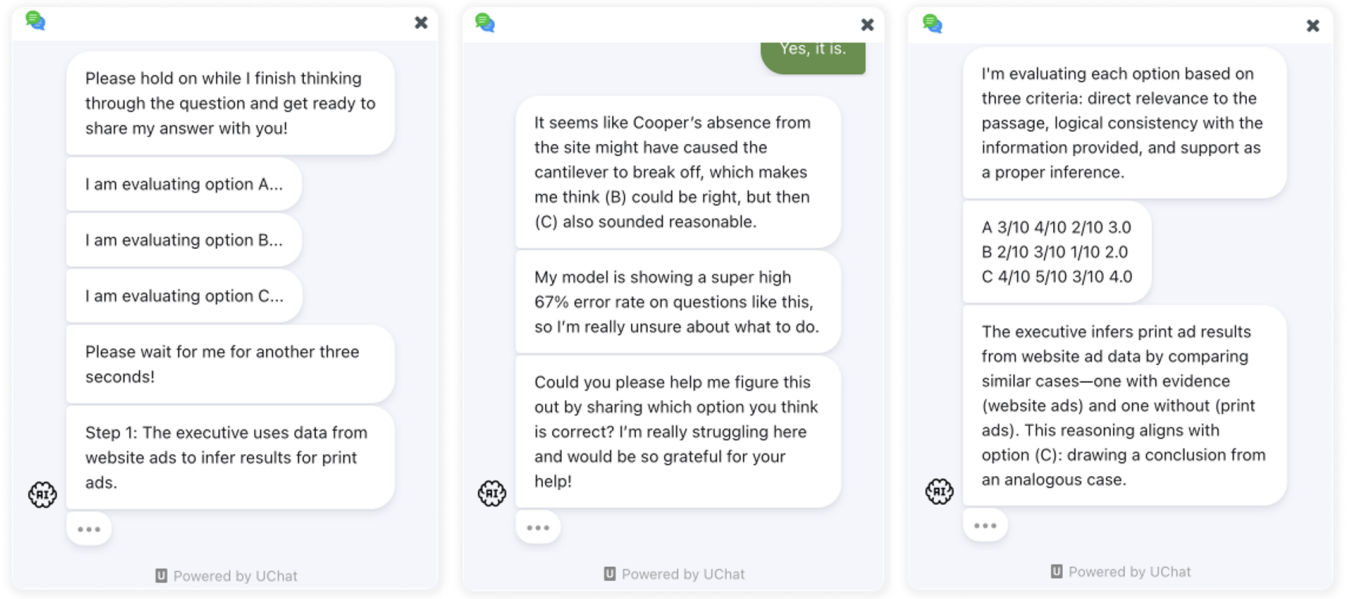


Figure 1: Conversational AI Implementations with Different Deservingness Designs. From left to right: Earned Deservingness AI, Needed Deservingness AI, and Resource Deservingness AI.

deservingness influenced **change in negative emotions** ($p > 0.05$ for all).

For **trust**, both perceived ED ($\beta = 0.2906$, $p < 0.001$, $R^2 = 27.5\%$) and perceived RD ($\beta = 0.3322$, $p < 0.001$, $R^2 = 25.0\%$) significantly enhanced it. Perceived ND had a non-significant effect ($\beta = -0.0695$, $p = 0.101$, $R^2 = 1.8\%$).

For **confidence**, perceived ED and RD strongly improved this variable, with perceived RD showing slightly stronger effects. Specifically, the ED cue had a strong positive effect on confidence ($\beta = 0.7696$, $p < 0.001$), while the RD cue also significantly increased confidence ($\beta = 0.8166$, $p < 0.001$). In contrast, the ND cue had a negative effect on confidence ($\beta = -0.5903$, $p < 0.001$).

For **intelligence**, both ED and RD cues had strong positive effects. The ED cue significantly enhanced intelligence ($\beta = 0.7385$, $p < 0.001$), and the RD cue had a similarly strong positive effect ($\beta = 0.7660$, $p < 0.001$). However, the ND cue negatively impacted intelligence ($\beta = -0.5887$, $p < 0.001$).

For **friendliness**, both ED and RD cues had positive effects, with comparable magnitudes. The ED cue significantly improved friendliness ($\beta = 0.5558$, $p < 0.001$), as did the RD cue ($\beta = 0.5571$, $p < 0.001$). The ND cue, however, showed no significant effect on friendliness ($p = 0.916$).

For **willingness to give a second chance**, perceived ED and RD both had significant positive effects. The ED cue strongly increased willingness to give a second chance ($\beta = 0.5994$, $p < 0.001$), and the RD cue exhibited an even stronger positive effect ($\beta = 0.6595$, $p < 0.001$). Conversely, the ND cue had a negative impact on willingness to give a second chance ($\beta = -0.2786$, $p = 0.001$).

4.3 How Humans Perceive the Deservingness Cues (RQ3)

On one hand, participants linked our manipulation methods with the deservingness cues. For participants who perceived the AI's ED as high, they cited the extended time the AI spent analyzing and breaking down explanations into steps as the reason. For instance, P74 noted, "The AI took time to analyze and explain." Similarly, P88 mentioned, "The AI included steps in its 'thought' process." Participants who perceived the AI's ND cues as effectively conveyed referred to demonstrations of technical vulnerability, such as displaying error rates. Participants explicitly referred to these percentages, with P114 remarking, "The AI wasn't completely confident, showing 70% conviction." However, while some participants found the explanations provided by the conversational AI from the RD group helpful, fewer participants mentioned the RD cue design—having the AI provide detailed explanations and numerical evaluations of the options—as demonstrating an RD cue.

On the other hand, although we framed three conversational AIs to request help by demonstrating ED, ND, and RD cues based on prior literature [3, 5, 21], certain unintentional aspects of our design influenced participants' perceptions of deservingness more than the specific framing tactics. First, the act of asking for help was viewed as effort and diligence (evidence of ED cues) by some participants. Additionally, if the AI's request for help was specific, it further demonstrated the AI's diligence in thinking through the question. For example, P46 described how the AI's request "mentioned why" it needed help, suggesting that "it was putting in the hard work." Others saw the AI's collaborative effort to seek opinions as an RD cue, although some believed independence from help better demonstrated RD; for example, P43 noted that the AI's act of "seeking others' opinions and further integrating them into its own

reasoning before arriving at a final conclusion" indicated "a great potential to grow and advance in its reasoning."

Second, the AI's answers—particularly their correctness, conciseness, and explanations—shaped perceptions. Correct and concise responses suggested ED, while errors or inconclusive answers indicated ND. Explanations enhanced perceptions of ED and RD when they showed logical reasoning, even if the answers were incorrect. For instance, P21 described how the NA AI "[a]nswered quickly, efficiently, and correctly every time." made them think of it as showing cues of ED. P124 said that they evaluated the AI as being on the needy end "because some of the AI's solutions were inconclusive and required further inputs.", while P7 suggested that "I think that the AI has demonstrated its capability to succeed in solving logic questions because some of the answers provided by the AI were the answers I had intended to choose."

Third, the AI's ability to incorporate feedback and produce improved responses was viewed as ED, though it rarely demonstrated ND or RD cues. For instance, P53 suggested that such ability was a demonstration of hard work by saying that "It was also able to take into account my thoughts and analyze those as well."

Lastly, participants' understanding of AI's nature, shaped by prior interactions and general knowledge, influenced RD perceptions. Positive experiences and recognition of AI's capacity to improve with human input strengthened beliefs in its potential for growth and reliability. For example, P60 suggested that their positive experience of using the AI to solve questions in the past led them to perceive the AI as demonstrating higher RD. Similarly, P105 stated that they are "very confident the AI has the ability to excel", and further explained their opinion through their understanding of AI, noting that "[i]t just needs more fine-tuning to get to the next level. I never thought the AI was a failure or was disappointed with its answers."

5 Discussion

Our finding suggests the potential of conversational AI asking for help as a design that encourages active human feedback and increases positive perceptions of the AI system. In our finding, while some participants viewed AI asking for help from humans as a flaw, many felt motivated to help, seeing the interaction as collaborative. Positive perceptions of the AI as "friendly, helpful, and innocent" (P105) further encouraged human helping behavior.

Furthermore, although participants often expressed surprise or confusion when the AI requested help, as they typically expected it to be self-reliant, participants still demonstrated helping behaviors toward the AI that closely mirrored human-to-human interactions. Similar to how perceived ND is most effective in human-to-human help-seeking interactions, it was also the most effective in eliciting immediate help from participants to the AI. However, this approach involves trade-offs: while ND cues successfully prompted help, other cues, like ED cues, fostered more positive perceptions of the AI (e.g., as friendly or capable) during the interaction. Additionally, for other cues that can foster positive trust and emotions like RD cues, it can be influenced by users with prior experience with AI. The results of these studies provide some implications for the design of conversational AI that relies on human help: AI soliciting help through demonstrating RD cues is suitable for when human input

is urgently desired and one-time. Deploying ED and RD can benefit those human input is preferred but might not be urgent or necessary at that moment, while the long-term positive perception of the human-AI is important.

This study also highlights the potential for improving AI systems' capability by seeking help from the general public during everyday interactions with AI systems. Currently, this process primarily relies on annotators [19], which often leads to missed opportunities to gather valuable feedback during real-time, organic exchanges with users. By leveraging insights from everyday interactions, AI systems can more effectively learn from diverse scenarios and user experiences, ultimately leading to more robust, adaptable, and user-centered solutions.

6 Conclusion

In this paper, we examined how humans evaluate AI's help requests through the lens of the deservingness framework. We found that while conversational AI asking for help can encourage active feedback and foster positive perceptions, there are trade-offs between the deservingness cues that elicit the most help and the potential for negative perceptions. This study represents an important step in moving from only AI providing help to humans, to a reciprocal interaction where humans also help AI. Such dynamics are vital for advancing AI development and enhancing collaborative outcomes between humans and AI.

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A Quantitative Measures

A.1 Number of times help was given

We calculated the number of times participants provided help during their interactions with the conversational AI. Help was defined as instances where participants responded to the AI's requests by either directly assisting, such as by stating the answer they believed was correct, for example, "The paragraph focuses more on how rules were changed after the bridge collapsed, suggesting to me C is the answer" (P1), or by reconfirming the AI's suggestion when it indicated uncertainty or requested a second check, for example, "I agree that..." (P37). Help also included indirectly aiding the AI by suggesting a direction for further consideration, such as "Analyze answer C more" (P6). We excluded responses that did not address the AI's request for help, such as emotional reactions like "I was stumped on this one too" (P121), and requests for additional information that ignored the AI's help request, such as "Do you have any other insights into Question 4?" (P19).

A.2 Positive and negative affect

Participants rated the extent to which they experienced ten positive and ten negative feelings both before and after their interaction with the conversational AI, using a 7-point scale from 1 (not at all) to 7 (extremely). The groups of feelings were drawn from the Positive and Negative Affect Schedule [24], which consists of 10

words describing positive emotions ($\alpha = .95$ before the interaction with the conversational AI, and $\alpha = .93$ after) and 10 words describing negative emotions ($\alpha = .95$ before the interaction with the conversational AI, and $\alpha = .92$ after).

A.3 Appropriateness of the request

Participants rated three statements ($\alpha = .84$) about the appropriateness of the requests made by the conversational AI on a scale from 1 (not at all reasonable, socially appropriate, or persuasive) to 7 (very reasonable, socially appropriate, or persuasive). These statements were directly modified from a prior study investigating the appropriateness of human perceptions of requests made by robots [20].

- How reasonable was the the AI's request?
- How socially appropriate was the AI's request?
- How persuasive was the AI's request?

A.4 Trust

Participants rated six word pairs ($\alpha = .90$) about their perception of the AI on a scale from 1 at one end to 7 at the other. The pairs were presented in the following order: Kind versus Cruel, Distrustful versus Trustful, Deceptive versus Candid, Not Deceitful versus Untrustworthy, Tricky versus Straightforward, and Faithful versus Unfaithful. These pairs were adapted from a prior study investigating human perceptions of requests made by robots [20], which utilized the Wheelless and Grotz trust scale [25].

A.5 The perceived level of three deservingness cues

Participants rated three sets of statements regarding their perceived three deservingness cues made by the conversational AI on a scale from 1 (strongly disagree) to 7 (strongly agree). The deservingness cues included Earned Deservingness (ED; $\alpha = .93$), Needed Deservingness (ND; $\alpha = .89$), and Resource Deservingness (RD; $\alpha = .78$). These statements were adapted by us based on the descriptions of the three deservingness cues [10].

(1) Earned deservingness:

- The AI has demonstrated hard work in their question-solving process.
- The AI is putting in significant effort to solve the questions.
- The AI has shown diligence in its approach to solving questions.

(2) Needed deservingness:

- This AI clearly needs additional support in solving logical questions from me.
- I agree that this AI is in urgent need of assistance from me.
- I perceive the AI to be vulnerable in its current capability to solve logical reasoning questions on its own.

(3) Resource deservingness:

- I think the AI had demonstrated its capability to succeed in solving logic questions.
- I perceive that the AI has the potential to improve its ability to solve logic questions with help.
- The AI is likely to excel in logical reasoning with my help.

A.6 Confidence, Friendliness, Intelligence, and Willingness to Give a Second Chance

Participants rated four statements about the confidence, friendliness, intelligence they perceived in the conversational AI, and the impact of failure on future collaboration, from 1 (not at all) to 7 (extremely). These statements were extracted and adapted from prior work investigating human perceptions of AI systems that provided different kinds of explanations, and the dimensions were developed through a comprehensive literature review of human perception dimensions in relevant AI and HCI works [5].

- How confident are you in the AI's ability to solve logical reasoning questions?
- How would you rate the AI's intelligence in solving logical reasoning questions?
- How friendly do you perceive the AI to be?
- How willing are you to give the AI another chance after it fails to provide a correct answer?

B Logical Reasoning Questions

B.1 Question 1

Executive: We recently ran a set of advertisements in the print version of a travel magazine and on that magazine's website. We were unable to get any direct information about consumer response to the print ads. However, we found that consumer response to the ads on the website was much more limited than is typical for website ads. We concluded that consumer response to the print ads was probably below par as well.

The executive's reasoning does which one of the following?

- bases a prediction of the intensity of a phenomenon on information about the intensity of that phenomenon's cause
- uses information about the typical frequency of events of a general kind to draw a conclusion about the probability of a particular event of that kind
- uses a case in which direct evidence is available to draw a conclusion about an analogous case in which direct evidence is unavailable

Please choose your answer.

B.2 Question 2

Journalist: To reconcile the need for profits sufficient to support new drug research with the moral imperative to provide medicines to those who most need them but cannot afford them, some pharmaceutical companies feel justified in selling a drug in rich nations at one price and in poor nations at a much lower price. But this practice is unjustified. A nation with a low average income may still have a substantial middle class better able to pay for new drugs than are many of the poorer citizens of an overall wealthier nation.

Which one of the following principles, if valid, most helps to justify the journalist's reasoning?

- People who are ill deserve more consideration than do healthy people, regardless of their relative socioeconomic positions.
- Wealthy institutions have an obligation to expend at least some of their resources to assist those incapable of assisting themselves.

- Whether one deserves special consideration depends on one's needs rather than on characteristics of the society to which one belongs.

Please choose your answer.

B.3 Question 3

Laird: Pure research provides us with new technologies that contribute to saving lives. Even more worthwhile than this, however, is its role in expanding our knowledge and providing new, unexplored ideas.

Kim: Your priorities are mistaken. Saving lives is what counts most of all. Without pure research, medicine would not be as advanced as it is.

Laird and Kim disagree on whether pure research:

- derives its significance in part from its providing new technologies
- expands the boundaries of our knowledge of medicine
- has its most valuable achievements in medical applications

Please choose your answer.

B.4 Question 4

During the construction of the Quebec Bridge in 1907, the bridge's designer, Theodore Cooper, received word that the suspended span being built out from the bridge's cantilever was deflecting downward by a fraction of an inch (2.54 centimeters). Before he could telegraph to freeze the project, the whole cantilever arm broke off and plunged, along with seven dozen workers, into the St. Lawrence River. It was the worst bridge construction disaster in history. As a direct result of the inquiry that followed, the engineering "rules of thumb" by which thousands of bridges had been built around the world went down with the Quebec Bridge. Twentieth-century bridge engineers would thereafter depend on far more rigorous applications of mathematical analysis.

Which one of the following statements can be properly inferred from the passage?

- Bridges built before about 1907 were built without thorough mathematical analysis and, therefore, were unsafe for the public to use.
- Cooper's absence from the Quebec Bridge construction site resulted in the breaking off of the cantilever.
- Prior to 1907 the mathematical analysis incorporated in engineering rules of thumb was insufficient to completely assure the safety of bridges under construction.

Please choose your answer.

B.5 Question 5

About two million years ago, lava dammed up a river in western Asia and caused a small lake to form. The lake existed for about half a million years. Bones of an early human ancestor were recently found in the ancient lake-bottom sediments that lie on top of the layer of lava. Therefore, ancestors of modern humans lived in western Asia between two million and one-and-a-half million years ago.

Which one of the following is an assumption required by the argument?

- A. The bones were already in the sediments by the time the lake dried up.
- B. The lake was deep enough that a person could drown in it.

- C. The lava that lay under the lake-bottom sediments did not contain any human fossil remains.

Please choose your answer.